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B25

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING																		
Program Name: B. Tech		Assignment Type: Lab	Academic Year: 2025-2026																	
Course Coordinator Name		Dr. Rishabh Mittal																		
Instructor(s) Name		<table border="1"> <tr><td>Mr. S Naresh Kumar</td></tr> <tr><td>Ms. B. Swathi</td></tr> <tr><td>Dr. Sasanko Shekhar Gantayat</td></tr> <tr><td>Mr. Md Sallauddin</td></tr> <tr><td>Dr. Mathivanan</td></tr> <tr><td>Mr. Y Srikanth</td></tr> <tr><td>Ms. N Shilpa</td></tr> <tr><td>Dr. Rishabh Mittal (Coordinator)</td></tr> <tr><td>Dr. R. Prashant Kumar</td></tr> <tr><td>Mr. Ankushavali MD</td></tr> <tr><td>Mr. B Viswanath</td></tr> <tr><td>Ms. Sujitha Reddy</td></tr> <tr><td>Ms. A. Anitha</td></tr> <tr><td>Ms. M.Madhuri</td></tr> <tr><td>Ms. Katherashala Swetha</td></tr> <tr><td>Ms. Velpula sumalatha</td></tr> <tr><td>Mr. Bingi Raju</td></tr> </table>		Mr. S Naresh Kumar	Ms. B. Swathi	Dr. Sasanko Shekhar Gantayat	Mr. Md Sallauddin	Dr. Mathivanan	Mr. Y Srikanth	Ms. N Shilpa	Dr. Rishabh Mittal (Coordinator)	Dr. R. Prashant Kumar	Mr. Ankushavali MD	Mr. B Viswanath	Ms. Sujitha Reddy	Ms. A. Anitha	Ms. M.Madhuri	Ms. Katherashala Swetha	Ms. Velpula sumalatha	Mr. Bingi Raju
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CourseCode	23CS002PC304	Course Title	AI Assisted Coding																	
Year/Sem	III/II	Regulation	R23																	
Date and Day of Assignment	Week9 – Wednesday	Time(s)	23CSBTB01 To 23CSBTB52																	
Duration	2 Hours	Applicable to Batches	All batches																	
Assignment Number: 17.3(Present assignment number)/24(Total number of assignments)																				
	Question		Expected Time																	

		<i>e to co mpl ete</i>
	Lab 17 – AI for Data Processing: Data Cleaning and Preprocessing Scripts Lab Objectives: • Learn how to clean raw datasets using AI-assisted Python scripting. • Apply preprocessing techniques such as handling missing values, encoding categorical data, and normalization. • Automate repetitive data-cleaning tasks with AI-generated code. • Understand how preprocessing impacts model performance.	
1	Task 1 Employee Dataset Cleaning Task: Use AI assistance to clean an employee information dataset. Instructions: • Handle missing values in salary, designation, and experience • Remove duplicate employee IDs • Standardize text values (job titles in lowercase) • Encode categorical features (designation, employment_type) Expected Output: A clean dataset suitable for HR analytics and prediction tasks. Prompt Generate Python (Pandas) code to clean an employee dataset. Tasks: - Handle missing values in salary, designation, experience - Remove duplicate employee IDs - Convert job titles to lowercase - Encode categorical columns (designation, employment_type) Also create a sample dataset and show output.	Week 9 - Wednesday

AI ASST 17.3_V.ipynb

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[] ▶ Start coding or generate with AI.

[]

```
import pandas as pd
import numpy as np

# Create a sample employee dataset
data = {
    'employee_id': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 3, 11],
    'name': ['Alice', 'Bob', 'Charlie', 'David', 'Eve', 'Frank', 'Grace', 'Heidi', 'Ivan', 'Judy', 'Charlie', 'Kevin'],
    'salary': [70000, 60000, np.nan, 80000, 75000, 62000, np.nan, 90000, 78000, 65000, 70000, 81000],
    'designation': ['Software Engineer', 'Data Analyst', 'Project Manager', 'Software Engineer', 'HR Manager', 'Data Analyst', np.nan, 'VP',
    'experience': [5, 3, 8, 6, 4, 2, np.nan, 10, 5, 7, 8, 6],
    'employment_type': ['Full-time', 'Contract', 'Full-time', 'Full-time', 'Part-time', 'Full-time', 'Full-time', 'Full-time', 'Contract', '']
}

df = pd.DataFrame(data)

print("Original DataFrame:")
display(df)
```

1. Handle Missing Values

- **Salary:** Impute with the median.
- **Designation:** Impute with 'Unknown'.
- **Experience:** Impute with the median.

[]

```
# Impute missing salary with the median
df['salary'] = df['salary'].fillna(df['salary'].median())

# Impute missing designation with 'Unknown'
```

Generate Python (Pandas) code to clean an employee dataset. Tasks: -
Handle missing values in salary, designation, experience - Remove duplicate
employee IDs - Convert job titles to lowercase - Encode categorical
columns (designation, employment_type) Also create a sample dataset and
show output.

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1. Handle Missing Values

- **Salary:** Impute with the median.
- **Designation:** Impute with 'Unknown'.
- **Experience:** Impute with the median.

```
[2]
✓ 0s
# Impute missing salary with the median
df['salary'] = df['salary'].fillna(df['salary'].median())

# Impute missing designation with 'Unknown'
df['designation'] = df['designation'].fillna('Unknown')

# Impute missing experience with the median
df['experience'] = df['experience'].fillna(df['experience'].median())

print("DataFrame after handling missing values:")
display(df)
```

DataFrame after handling missing values:

	employee_id	name	salary	designation	experience
0	1	Alice	70000.0	Software Engineer	
1	2	Bob	60000.0	Data Analyst	
2	3	Charlie	72500.0	Project Manager	
3	4	David	80000.0	Software Engineer	
4	5	Eve	75000.0	HR Manager	
5	6	Frank	62000.0	Data Analyst	
6	7	Grace	72500.0	Unknown	

Generate Python (Pandas) code to clean an employee dataset. Tasks: - Handle missing values in salary, designation, experience - Remove duplicate employee IDs - Convert job titles to lowercase - Encode categorical columns (designation, employment_type) Also create a sample dataset and show output.

◆ Sure, I can help with that! Here's the Python (Pandas) code to perform the data cleaning tasks you described, starting with a sample dataset creation.

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Explanation

- Missing salary → replaced with **mean**
- Experience → replaced with **median**
- Duplicate IDs → removed using `drop_duplicates`
- Text standardized using `.str.lower()`
- Categorical encoding → `get_dummies()`

Task 2 – Student Performance Data Preprocessing

Task:

Preprocess student performance data using AI assistance.

Instructions:

- Convert exam_date column into datetime format
- Extract new features (month, semester, day)
- Normalize total_marks
- Detect and handle outliers in marks using IQR or Z-score

Expected Output:

A transformed dataset with time-based academic features and normalized marks.

Prompt

Write Python code using Pandas to preprocess student data.

Tasks:

- Convert exam_date to datetime
- Extract month, semester, day
- Normalize total_marks
- Detect outliers using IQR and remove them

Include sample data.

AI ASST 173_Viptynb

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```
[ ]
import pandas as pd
import numpy as np
from sklearn.preprocessing import MinMaxScaler

# Create a sample student dataset
data = {
    'student_id': [101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115],
    'name': ['Alex', 'Brian', 'Cathy', 'David', 'Eve', 'Frank', 'Grace', 'Heidi', 'Ivan', 'Judy', 'Liam', 'Mia', 'Noah', 'Olivia', 'Peter'],
    'exam_date': ['2023-09-15', '2023-09-15', '2023-10-20', '2023-10-20', '2023-11-01',
                 '2023-11-01', '2024-02-10', '2024-02-10', '2024-03-05', '2024-03-05',
                 '2023-04-10', '2023-04-10', '2024-01-20', '2024-01-20', '2023-09-15'],
    'subject': ['Math', 'Science', 'Math', 'English', 'Science',
               'English', 'Math', 'Science', 'English', 'Math',
               'Science', 'Math', 'English', 'Science', 'Math'],
    'total_marks': [85, 92, 78, 65, 95, 70, 88, 50, 99, 81, 10, 93, 75, 80, 200]
}
df_students = pd.DataFrame(data)

print("Original Student DataFrame:")
display(df_students)
```

1. Convert exam_date to Datetime and Extract Features

- Convert the exam_date column to datetime objects.
- Extract the month as a number.
- Define a simplified 'semester' based on months (e.g., Spring).
- Extract the day of the month.

```
[ ]
# Convert 'exam_date' to datetime objects
df_students['exam_date'] = pd.to_datetime(df_students['exam_date'])
```

The data cleaning process has been fully completed according to your instructions. If you have any further questions or need additional analysis on the cleaned dataset, feel free to ask!

Promot Write Python code using Pandas to preprocess student data. Tasks:

▶ Accept & Run ✓ Accept ✕ Cancel

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```
display(df_students_cleaned)
```

Identified Outliers in 'total_marks':

student_id	name	exam_date	subject	total_marks	exam_month	exam_day	exam_semester	normalized_marks	
10	111	Liam	2023-04-10	Science	10	4	10	Spring	0.0
14	115	Peter	2023-09-15	Math	200	9	15	Fall	1.0

DataFrame after removing outliers from 'total_marks':

student_id	name	exam_date	subject	total_marks	exam_month	exam_day	exam_semester	normalized_marks	
0	101	Alex	2023-09-15	Math	85	9	15	Fall	0.394737
1	102	Brian	2023-09-15	Science	92	9	15	Fall	0.431579
2	103	Cathy	2023-10-20	Math	78	10	20	Fall	0.357895
3	104	David	2023-10-20	English	65	10	20	Fall	0.289474
4	105	Eve	2023-11-01	Science	95	11	1	Fall	0.447368
5	106	Frank	2023-11-01	English	70	11	1	Fall	0.311774
6	107	Grace	2024-02-10	Math	88	2	10	Spring	0.407692
7	108	Heidi	2024-02-10	Science	50	2	10	Spring	0.162171
8	109	Ivan	2024-03-05	English	99	3	5	Spring	0.521739
9	110	Judy	2024-03-05	Math	81	3	5	Spring	0.375000
11	112	Mia	2023-04-10	Math	93	4	10	Spring	0.460526
12	113	Noah	2024-01-20	English	75	1	20	Spring	0.260870
13	114	Olivia	2024-01-20	Science	80	1	20	Spring	0.289474

Next steps: [Generate code with outliers](#) [New interactive sheet](#)

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Explanation

- Date converted → enables feature extraction
- Semester logic → based on month
- Normalization → Min-Max scaling
- Outliers → removed using IQR method

Task 3 – Student Health Data Preparation

Task:

Clean and preprocess student health records using AI scripts.

Instructions:

- Handle missing values in height, weight, BMI
- Encode categorical features (medical_condition, fitness_status)
- Scale numerical features using standard scaling
- Split dataset into training and testing sets

Expected Output:

A preprocessed dataset ready for predictive health analysis.

Prompt

Generate Python code to preprocess student health dataset.

Tasks:

- Handle missing values (height, weight, BMI)
- Encode categorical (medical_condition, fitness_status)
- Apply standard scaling
- Split into train and test sets

AI ASST 17.3_V.ipynb

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[] Start coding or generate with AI.

[10] 0s

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split

# Create a sample student health dataset
data = {
    'student_id': range(1, 16),
    'age': [16, 17, 16, 18, 17, 16, 18, 17, 16, 18, 17, 16, 18, 17, 16],
    'gender': ['Male', 'Female', 'Male', 'Female', 'Male', 'Female', 'Male', 'Female', 'Male', 'Female', 'Male', 'Female', 'Male', 'Female', 'Male'],
    'height_cm': [170, 165, np.nan, 175, 168, 162, 180, 170, 172, 166, 178, 160, np.nan, 169, 171],
    'weight_kg': [65, 58, 68, np.nan, 60, 55, 75, 62, np.nan, 59, 70, 53, 72, 61, 64],
    'bmi': [22.5, 21.3, 23.5, 22.8, np.nan, 20.9, 23.1, 21.5, 22.0, 21.4, 21.9, np.nan, 23.0, 21.2, 22.3],
    'medical_condition': ['None', 'Asthma', 'None', 'Allergy', np.nan, 'None', 'Asthma', 'None', 'Allergy', 'None', 'None', 'Asthma', 'None', np.nan, 'None'],
    'fitness_status': ['Good', 'Excellent', 'Fair', 'Good', 'Fair', 'Good', 'Fair', 'Good', 'Fair', 'Good', 'Fair', 'Good', 'Fair', 'Good', 'Fair']
}

df_health = pd.DataFrame(data)

print("Original Student Health DataFrame:")
display(df_health)
```

Original Student Health DataFrame:

	student_id	age	gender	height_cm	weight_kg	bmi
0	1	16	Male	170.0	65.0	22.5
1	2	17	Female	165.0	58.0	21.3
2	3	16	Male	NaN	68.0	23.5
3	4	18	Female	175.0	NaN	22.8

Prompt Generate Python code to preprocess student health dataset. Tasks:

- Handle missing values (height, weight, BMI) - Encode categorical (medical_condition, fitness_status) - Apply standard scaling - Split into train and test sets

Okay, I'll generate the Python code using Pandas and Scikit-learn to preprocess a student health dataset based on your requirements. This will

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Explanation

- Missing values → replaced with mean
- Encoding → one-hot encoding
- Scaling → StandardScaler (important for ML)
- Splitting → training/testing dataset

Task 4: Real-Time Application: Ride-Sharing Data Cleaning

Scenario:

A ride-sharing company has trip data with errors.

Requirements:

- Standardize pickup and drop locations
- Fill missing fare amounts with median values
- Remove duplicate ride entries
- Normalize driver ratings
- Generate a summary of data improvements

Expected Output:

A clean dataset ready for operational analysis.

Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.

Prompt

Write Python code to clean ride-sharing dataset.

Tasks:

- Standardize pickup/drop locations
- Fill missing fare using median
- Remove duplicates
- Normalize driver ratings
- Show summary of improvements

AI ASST 173_V.ipynb

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```
[ ]
import pandas as pd
import numpy as np
from sklearn.preprocessing import MinMaxScaler

# Create a sample ride-sharing dataset
data = {
    'ride_id': [101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 101, 111, 112, 113, 114],
    'pickup_location': ['Downtown ', 'airport', 'Suburb A', 'downtown', ' Airport', 'Suburb B', 'downtown', 'airport ', 'Suburb A', 'Downtown', 'airport', 'Suburb B', 'Downtown', 'Suburb A', 'Suburb B'],
    'dropoff_location': ['Airport', 'Downtown', 'Suburb B', 'Airport', 'Suburb A', 'Downtown', 'Suburb B', 'Downtown', 'Airport', 'Suburb B', 'Downtown', 'Suburb A', 'Downtown', 'Suburb B', 'Suburb A'],
    'distance_km': [10.5, 25.3, 15.0, 11.2, 24.8, 14.5, 12.0, 26.0, 16.0, 18.0, 10.5, 20.0, 13.0, 19.5, 14.5],
    'fare_usd': [20.0, 45.0, 28.0, np.nan, 42.0, 27.0, 22.0, 47.0, np.nan, 35.0, 20.0, 38.0, 25.0, 36.0, 27.0],
    'driver_rating': [4.8, 3.5, 4.9, 4.2, 5.0, 3.0, 4.5, 4.0, 4.7, 3.8, 4.8, 4.6, 4.1, 3.9, 3.0],
    'passenger_count': [1, 2, 1, 1, 3, 2, 1, 1, 2, 1, 1, 2, 1, 2, 2]
}

df_rides = pd.DataFrame(data)

print("Original Ride-Sharing DataFrame:")
display(df_rides)
```

1. Standardize Pickup/Drop Locations

We will convert the pickup_location and dropoff_location to lowercase and strip whitespace for consistency.

```
[ ]
# Standardize location names (lowercase and strip whitespace)
df_rides['pickup_location'] = df_rides['pickup_location'].str.lower().strip()
df_rides['dropoff_location'] = df_rides['dropoff_location'].str.lower().strip()

print("DataFrame after standardizing locations:")
display(df_rides)
```

Prompt Write Python code to clean ride-sharing dataset. Tasks: - Standardize pickup/drop locations - Fill missing fare using median - Remove duplicates - Normalize driver ratings - Show summary of improvements

Sure, I can help you with cleaning a ride-sharing dataset. I'll generate the Python code using Pandas to perform the tasks you've outlined, including:

Accept & Run Accept Cancel

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